Student performance dashboard analysis



Internship Project Presentation

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📌 Project Overview

- **Purpose:** To analyze student performance data and identify factors influencing academic success.
- **Scope:** Covers variables such as study hours, attendance, parental education, internet access, and subject scores.
- **Objective:** Provide actionable insights and recommendations for educators through data analysis.

Approach: Applied data cleaning, exploratory data analysis (EDA), and visualization techniques using Python and Power BI.

- **Outcome:** Developed an interactive Power BI dashboard that highlights trends, correlations, and key metrics for decision-making.



Problem Statement

- Educational institutions struggle to identify the key factors that influence student performance.
- Large datasets make it difficult to pinpoint which variables (study hours, attendance, parental education, internet access, etc.) have the most impact.
- Without proper analysis, valuable insights remain hidden and decision-making becomes guesswork.
- Lack of actionable insights prevents educators from designing effective strategies to improve learning outcomes.
- This project addresses the challenge by applying data cleaning, exploratory analysis, and visualization techniques to uncover meaningful patterns.

Dataset Description

- Contains student records with attributes such as study hours, subject scores, attendance percentage, parental education level, and internet access.
- Each row represents one student; each column represents a specific variable.
- Provides both numerical (scores, hours, attendance) and categorical (gender, parental education, internet access) data.
- Serves as the foundation for cleaning, analysis, and visualization.



Tools Used

Power BI

Python

SQL

GitHub

Data Cleaning Process

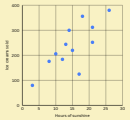
- Inspection – Reviewed dataset for missing values, duplicates, and incorrect data types.
- Handling Missing Values – Imputed averages for numeric fields (study hours, attendance) and labeled unknowns for categorical fields (parental education).
- Duplicate Removal – Eliminated repeated student records to ensure accuracy.
- Data Type Correction – Converted scores and study hours into numeric format; standardized categorical fields (e.g., “Male” vs “male”).
- Column Renaming & Transformation – Renamed columns for clarity, normalized attendance into percentage format, and standardized study hours.
- Outcome – Dataset became clean, consistent, and ready for analysis and visualization.





Data Visualizations





The project includes a variety of visualizations designed to highlight important aspects of student performance:

- Scatter Plot – Shows the relationship between study hours and subject scores, revealing a positive correlation.
- Line Graph – Tracks performance trends over time, helping identify improvements or declines.
- Bar Chart – Compares average scores across categories such as gender and internet access.
- Pie Chart – Illustrates the distribution of parental education levels among students.
- KPI Cards – Summarize key metrics like average study hours, attendance percentage, and pass rate.

STUDENT PERFORMANCE DASHBOARD

88.24
PassRate

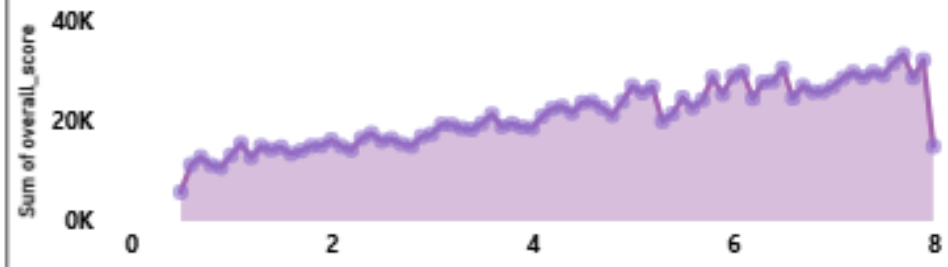
25K
Count of Students

4.25
AvgStudyHours

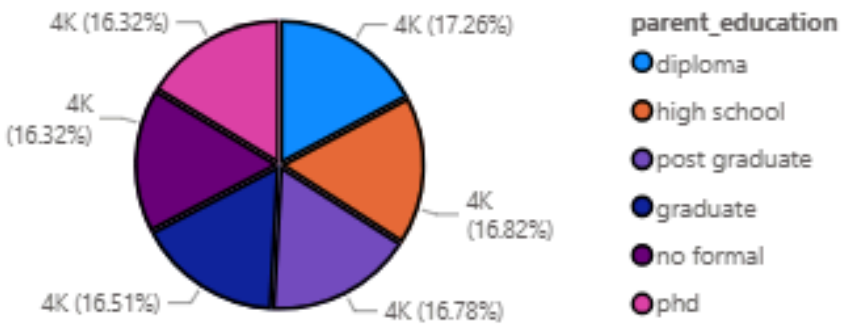
Attendance_Percentage
All

Gender
All

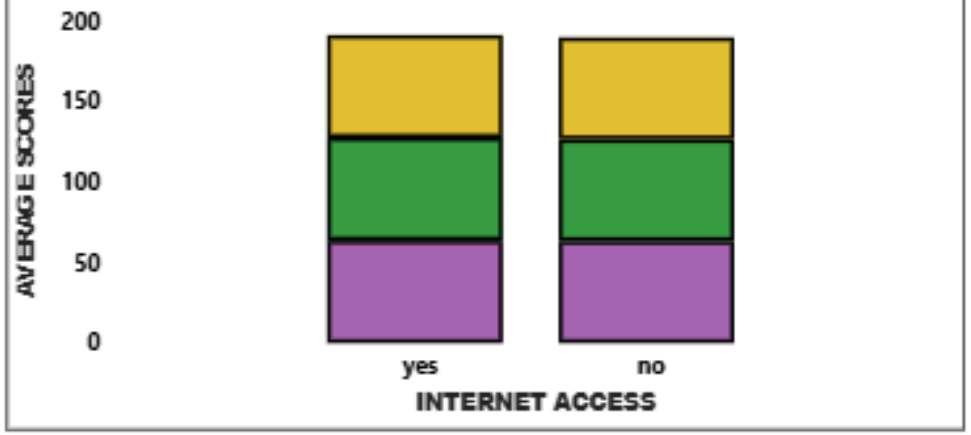
Average score vs Study hours



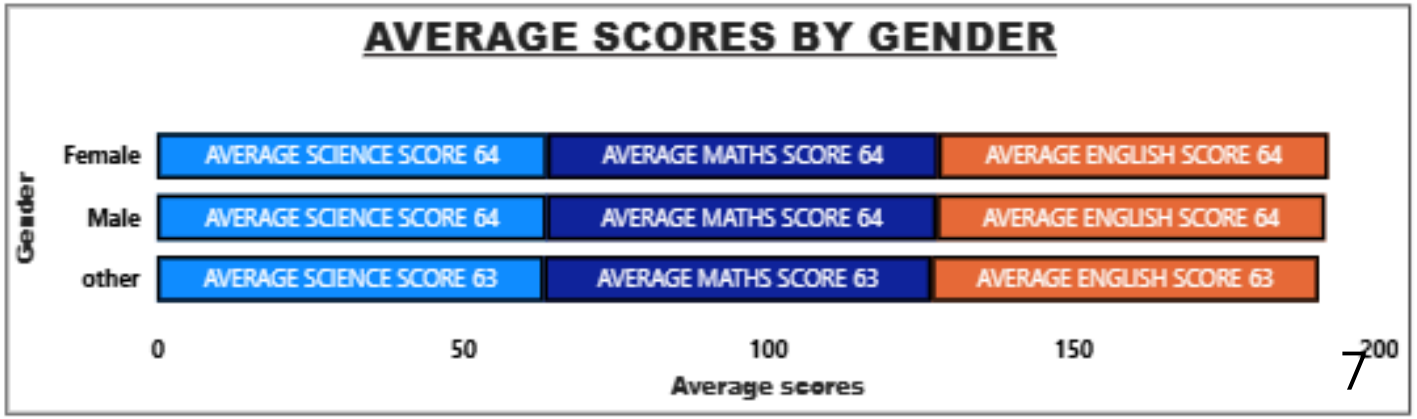
Students by Parent_Education



AVERAGE by INTERNET ACCESS



AVERAGE SCORES BY GENDER





Key Insights

-Study Hours and Scores – Students who dedicate more study hours consistently achieve higher average scores across subjects, showing a strong positive correlation.

-Attendance Impact – Higher attendance percentages directly contribute to better performance, highlighting the importance of consistent classroom participation.

-Parental Education Influence – Students with parents holding higher education levels tend to perform better, suggesting a link between family background and academic outcomes.

-Internet Access Advantage – Availability of internet access provides students with a slight edge in performance, likely due to access to digital learning resources.

Conclusion



- The project successfully analyzed student performance data using Python, SQL, and Power BI.
- Data cleaning ensured accuracy and consistency, preparing the dataset for meaningful insights.
- Exploratory Data Analysis (EDA) uncovered correlations between study hours, attendance, parental education, and performance.
- Interactive Power BI dashboard provided clear visualizations and actionable insights for educators.
- The findings can guide institutions in improving learning outcomes through data-driven strategies.

Thank You

- Thank you for reviewing this project.
- I appreciate the opportunity to present my work.
- Looking forward to applying these skills in future projects and real-world scenarios.



